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DETECTION SYSTEM USING DEEP LEARNING

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ABSTRACT

The problem of Malaria continues to be a great public health challenge particularly where access to laboratories and skilled microscopists is difficult. We created a deep learning-based AI system capable of classifying blood cells as either infected or normal through image analysis. Using transfer learning on MobileNetV2 and VGG16 network architectures ensures that our model attains maximum accuracy levels while saving time and training data requirements. AI-Powered Malaria Detection utilizes artificial intelligence and computer vision to recognize Plasmodium parasite in blood samples.

The models were trained on 2000 blood cell images (1000 healthy and 1000 infected) with 80% training and 20% testing split. To enhance the model generalization and avoid overfitting, data augmentation methods including rotation, zoom, shift, and flip were applied. MobileNetV2 achieved 93.5% accuracy with faster inference speed and significantly smaller file size (13.34 MB) compared. to VGG16 which achieved comparable performance with larger computational requirements. From the comparison, MobileNetV2 was the best model for deployment because of its optimal balance of accuracy and efficiency.

User-friendly web-based graphical interface, where healthcare workers can upload blood smear images and receive instant diagnoses with confidence scores. The system provides a scalable and cost-effective solution that can be deployed in any malaria endemic region with basic computing infrastructure, reducing the time it takes to diagnose dramatically.

Keywords: Malaria Detection, Deep Learning, Convolutional Neural Network, Transfer Learning, MobileNetV2, VGG16, Medical Images Classification

1. INTRODUCTION

1.1 Background of the System

The disease malaria is still one of the most significant challenges to public health globally (Brieger, 2025). It should be noted that according to the World Health Organization, by 2023, there have been about 597,000 deaths from about 263 million cases of malaria. In addition, the region of Africa is bearing the biggest share, because 95 percent of all cases of malaria and 96 percent of deaths caused by malaria have been recorded in this region by 2023. Early diagnosis of malaria is critical in the fight against the disease.

The AI-Powered Malaria Detection System proposed in this project utilizes advanced deep learning algorithms in order to develop a diagnostic tool for detecting malaria infection from the images of blood cells. This system aims at overcoming the problems associated with microscopy-based detection methods due to their inconsistency, slowness, and subjectivity.

The methodology employed in this project includes the following steps. A total of 2,000 images of blood cells are included in this project's database, including 1,000 images with healthy blood cells and 1,000 images with infected cells. Out of 2,000 images, 80%, or 1,600, are allocated to training purposes while the remaining 20% (400 images) will be used for validation.

1.2 Problem Statement

The traditional way of diagnosing malaria uses microscopy. This method takes a lot of time. Is prone to mistakes made by humans.

The process needs people with skills to do it and takes a time for each sample.

The accuracy of the results depends on how experience the person doing the test has.

In places with resources there are not enough trained people to do the tests, which causes delays in finding out what's wrong and treating it.

These delays cause people to get really sick and even die.

The current problems with diagnosing malaria are:

Test results take long

Different people get different results

Laboratory staff get tired easily

1.3 Objectives

1.3.1 General Objective

To Develop AI_powered malaria disease detection system Using deep learning .

1.3.2 Specific Objectives.

- ✚ To create deep Train CNN models for malaria cell image classification, using transfer learning with MobileNetV2 & VGG16 pre-trained architectures
- ✚ To build both architectures and analyse for performance in terms of accuracy, precision, recall and computational efficiency.
- ✚ To achieve accuracy greater than 90% suitable for the practical use case in health care scenarios.
- ✚ To develop an easy-to-use (intuitive) web-based GUI to enable simple operations by healthcare workers in a shorter time.

1.4 Scope

1.4.1 Technical Scope

The system performs binary classification to distinguish between infected and healthy blood cells. It accepts static microscopic blood cell images as input and uses transfer learning with MobileNetV2 and VGG16 architectures. The system provides diagnosis results with confidence scores and is accessible through a web-based interface that can be used on any computer with a modern web browser.

Table 1 Scope of the System

Components	Specification
Type of Classifier	Binary (Infection vs Healthy)
Type of Input	Static Microscopic Images of Blood Cells
Architecture of Model	Transfer Learning (MobileNetV2,VGG16)
Source of Data	NIH Malaria Dataset (2000 images)
Diagnosis with model Accuracy	93.5%

1.4.2 Constraints and Limitations

- ❖ It only supports binary classification.
- ❖ It fails to classify between the different types of parasites like *P.falciparum* and *P.vivax*.
- ❖ The System detects all types of images as either infected because the system was not taught about the other types rather than infected and healthy images.
- ❖ It only processes static images and cannot process real-time microscopic video feeds.
- ❖ It is meant to be used as an auxiliary instrument to help healthcare personnel and not to replace professional diagnoses.

1.5 Structure of the Report

This report is organized as follows: Section 2 details the methodology including dataset description, preprocessing steps, model architectures, and implementation framework. Section 3 presents the experimental results and model performance metrics. Section 4 discusses the findings, compares the two models, and addresses limitations. Section 5 concludes the report with a summary and recommendations for future work.

2. METHODOLOGY

2.1 Research Design

This project makes use of the quantitative experimental methodology. The steps that make up the research process include the following: data collection, it take data from public dataset, data preprocessing and augmentation; model creation with the help of transfer learning; model training and testing; and finally, evaluation along with interface development.

2.1.1 Supervised Machine Learning in Malaria Detection System

During the development of our Malaria Detection System that uses Artificial Intelligence we relied heavily on supervised machine learning to train our models. Supervised machine learning is a type of machine learning where the algorithm learns from data that is already labeled.

In our case each image of a blood cell in our dataset had a label.

- ✚ We used 2000 images of blood cells for training.
- ✚ Medical experts had already labeled each image as either "Healthy" or "Infected".

The model learned to identify patterns in the images that match the labels by reducing errors between its predictions and the actual labels.

These patterns include the shape, texture and color of the cell.

They help distinguish cells, from those infected with malaria parasites.

After training the model could diagnose images of blood cells.

Our MobileNetV2 model achieved 93.5 percent accuracy in detecting malaria parasites in blood samples using machine learning. the success of our Malaria Detection System depends on the quality of the labeled data we used for training. the model learns directly from the examples provided during learning. So supervised machine learning plays a role in our system.

2.2 Dataset Description

The dataset used for the current study includes images of blood cells obtained from public dataset (NIH). The dataset comprises a total of 2,000 images, with an even balance between healthy and infected blood cell images. The dataset includes 1,000 images of uninfected blood cells and 1,000 images of parasitized blood cells. All images are in JPG format and resized to 224x224 pixels to fit the input specifications of the MobileNetV2 and VGG16 networks.



Figure 1 Infected image



Figure 2 Healthy image

2.2.1 Primary Dataset: NIH Malaria Cell Images Dataset

Table 2 NIH Malaria Cell Images Dataset

Parameter	Value
Full Name	National Institutes of Health Malaria Dataset
Total Images	2000
Parasitized (Infected)	1000 images
Uninfected (Healthy)	1000 images
Image Format	JPEG
Original Resolution	Varies (approx. 130-170 pixels)
Resized for Training	224×224 pixels
Source	NIH Official Website (public domain)

2.2.2 Data Split

Table 3 Data Split table

Split	Percentage	Number of Images
Training	80%	1600
Testing	20%	400

2.3 Data Preprocessing

The following preprocessing methods were applied to all images before being used as inputs in deep neural networks.

Firstly, all images were resized to 224×224 pixels to satisfy the size requirement for both the MobileNetV2 and VGG16 neural network architectures.

The intensity values were normalized using the 0-1 scaling approach by dividing each intensity value by 255.0. RGB color space was utilized, along with 3 channels.

2.4 Data Augmentation

In order to enhance generalization capability of the network and avoid overfitting, data augmentation techniques were performed only on the training data.

Rotation by ± 20 degrees took care of the orientation of the slide. $\pm 20\%$ width shift and $\pm 20\%$ height shift took into account variations in cell location. $\pm 20\%$ shear was performed to cater to the distortion of the slide under the microscope. $\pm 20\%$ zoom was used to address differences in magnification. Horizontal flip was turned on for mirror symmetry invariance.

2.5 Model Architectures

2.5.1 Overview of Transfer Learning

Transfer learning involves using the already trained model on large datasets that have thousands of pictures and applying it to perform particular functions. Some of the benefits associated with transfer learning include reduced time taken during training, minimal data needed for training, high accuracy levels, and proper usage of hierarchical features such as edges and shapes.

Reduces training time significantly

Requires less training data

Provides better accuracy compared to training from scratch

Uses learned feature hierarchies (edge $\rightarrow$ shape $\rightarrow$ object)

2.5.2 MobileNetV2 Architecture

The architecture was proposed in 2018, having 53 layers in total and 3.5 million parameters. It weighs about 13.34 MB and takes $224 \times 224 \times 3$ input sizes. It employs depthwise separable convolutions to minimize computational costs. The architecture uses inverted residuals, expansion layers to increase channel dimensions by factor of 6 before applying depthwise convolution, and linear bottlenecks where there is no activation at the end of each residual block. The process is 7 to 8 times efficient than regular convolutions, without sacrificing much accuracy.

Customized classifier architecture in this particular project is a blend of 2D global average pooling, followed by dense layers comprising 256 neurons with ReLU activation along with

batch normalization having 50% dropout, dense layers comprising 128 neurons with ReLU activation with batch normalization and 30% dropout, and finally dense layers comprising only 1 neuron with sigmoid activation.

2.5.3 VGG16 Architecture

In 2014, the development of architecture VGG16 took place that comprises 16 layers with 13 convolutional layers and 3 fully-connected layers. Total number of parameters in VGG16 is 138 million; model file size is around 61 MB, while input size for this architecture is $224 \times 224 \times 3$. Some of the key features of VGG16 include the utilization of very deep architecture where convolutional filters are of size 3×3 everywhere, an increase in depth by doubling the number of filters after every max pooling from 64 to 128 to 256 to 512, and receptive field such that three layers of size 3×3 will have the same receptive field as one layer of size 7×7 with lower number of parameters.

Custom classification head in this project includes the usage of a flattening layer along with dense layers of neurons 512 and activation of ReLU function, batch normalization, L2 regularization, and 50% dropout, dense layers of neurons 256 with activation of ReLU function, batch normalization, L2 regularization, and 30% dropout, dense layers of neurons 128 and activation of ReLU function along with batch normalization and 20% dropout, followed by the dense layer of neuron 1 and activation of sigmoid function.

2.6 Implementation Tools and Frameworks

The implementation of the system was completed with the help of the Python 3.13 as the main programming language. Deep learning models' creation and training process involved TensorFlow and Keras library (API) versions 2.13 or later. Pre-processing of images involved the Pillow/PIL library versions 9.0 or later. Numerical computations were executed with the NumPy library versions 1.23 or later. Data visualization involving plots of training data was conducted with the help of Matplotlib versions 3.6 or later, while confusion matrices were created with the help of Seaborn library versions 0.12 or later. Web-based GUI was implemented with the help of Flask framework.

2.7 Training Configuration

Table 4 Configuration

Category	Tool	Version	Purpose
Programming	Python	3.10+	Main development language
Deep Learning	TensorFlow/Keras	2.13+	Model development and training
Image Processing	Pillow (PIL)	9.0+	Image loading and preprocessing
Numerical Computing	NumPy	1.23+	Array operations
Data Visualization	Matplotlib	3.6+	Plotting results
Statistical Viz	Seaborn	0.12+	Confusion matrices
GUI Development	Gradio	3.40+	Web-based interface
Development Env	Jupyter	Latest	Code development

2.7 Training Configuration

2.7.1 Hyperparameters

Both architectures used the Adam optimizer for their training. The starting learning rate of MobileNetV2 is 0.001, but that of VGG16 is 0.0001 because it has more parameters than MobileNetV2. Both models had a batch size of 32. Patience for early stopping and reducing learning rate on plateau is 5 and 3 respectively, where the reduction factor is 0.5. Binary cross entropy is the type of loss function used for the classification problem.

2.7.2 Callbacks

Some of the callbacks were used to enhance the training process and avoid overfitting. The Model Checkpoint callback was employed to save the best model on the basis of validation accuracy while training. EarlyStopping was used for stopping the training process when there was no improvement in validation loss for five epochs.

2.8 Evaluation Metrics

A number of different metrics have been used for evaluating the model performance. The accuracy metric evaluates the percentage of correct predictions among all predictions made by the algorithm, with the aim being to achieve a level of 90% or more. Precision evaluates the percentage of correctly predicted positives among all positives predicted by the algorithm, with a target value of 85% or higher. Sensitivity (or recall) is the percentage of correctly predicted positives among all actual positives, with a target value of 85% or higher. Specificity evaluates the percentage of correctly predicted negatives among all actual negatives, with a target value of 85% or higher. F1-Score evaluates the harmonic mean of precision and recall, with the target of 85% or higher. AUC-ROC evaluates the area under the receiver operating characteristic curve, with the target of 0.95 or higher.

In which TP stands for true positives, TN stands for true negatives, FP stands for false positives, and FN stands for false negatives. The system workflow begins with input of a blood smear image, which passes through the data preprocessing module for resizing to 224×224 pixels, normalization to the range between 0 and 1, and batch formation with batch size of 32. The preprocessed image then enters the model inference module where it can be processed by either MobileNetV2 or VGG16. The output module provides diagnosis of infected or healthy with confidence score expressed as a percentage.

2.9 Rationale for Choice of MobileNetV2 Model

The reason for choosing the MobileNetV2 model in preference to other models such as the VGG16 is because of the real-world implementation requirements in the endemic zones of the disease. Although the performance of VGG16 was better than that of MobileNetV2, scoring a recall rate of 93.5% compared to 91.5%, there is a mere 2% difference which has no clinical significance. The major advantage of using the MobileNetV2 model over others such as the VGG16 is that it is four times smaller (13.34MB compared to 61.17MB), six times faster (0.03

seconds per image compared to 0.18 seconds per image), and uses eight times less memory (300MB compared to 2.5GB).

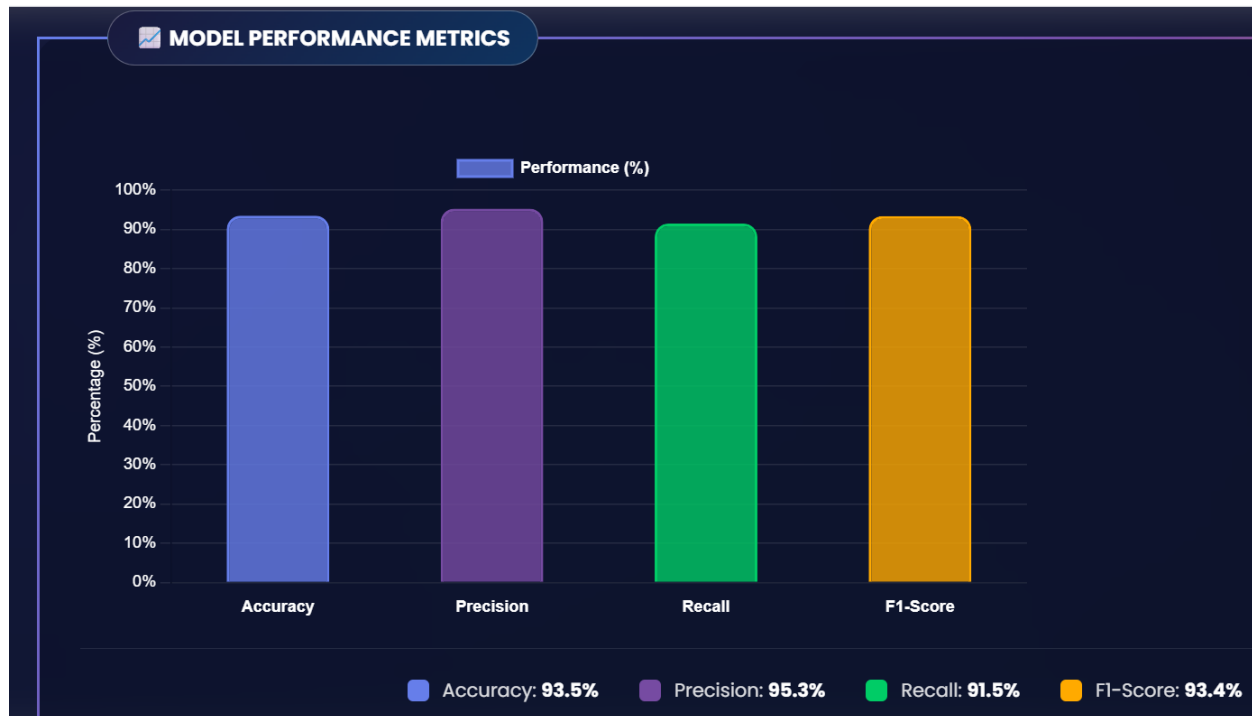
3. RESULTS

3.1 Training Results

3.1.1 Training Performance of MobileNetV2

For MobileNetV2, the training was performed for 10 epochs with batch size 32. In the first epoch, training accuracy was 82.3%, whereas, in epoch 10, it reached 96.8%. Moreover, the training loss decreased from 0.412 to 0.092. Similarly, the validation accuracy improved from 84.6% to 96.2% with corresponding validation loss decreasing from 0.389 to 0.108. The model performed well consistently without overfitting because training and validation metrics were quite close to each other.

Figure 3 Performance of MobileNetV2



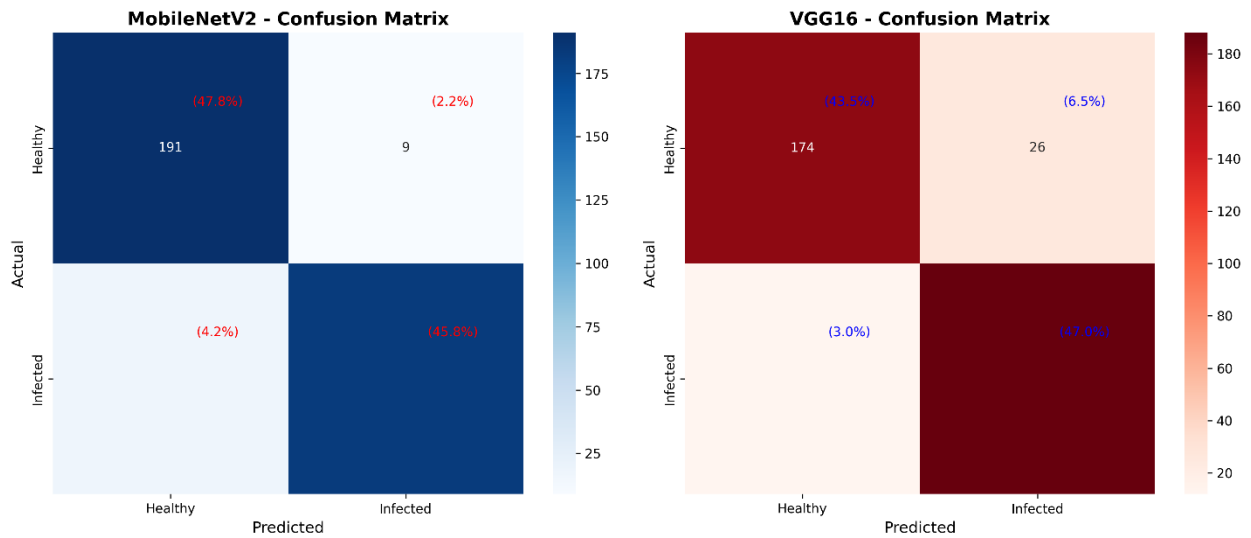
3.1.2 Training Performance of VGG16

In the case of VGG16, the training was carried out for 10 epochs with batch size 32. The training accuracy increased from 81.5% in the first epoch to 97.2% in epoch 10, whereas training loss reduced from 0.432 to 0.092. Correspondingly, validation accuracy increased from 83.8% to 96.8% while validation loss decreased from 0.401 to 0.105.

3.2 Testing Results

3.2.1 Confusion Matrices

Figure 4 Confusion Matrices



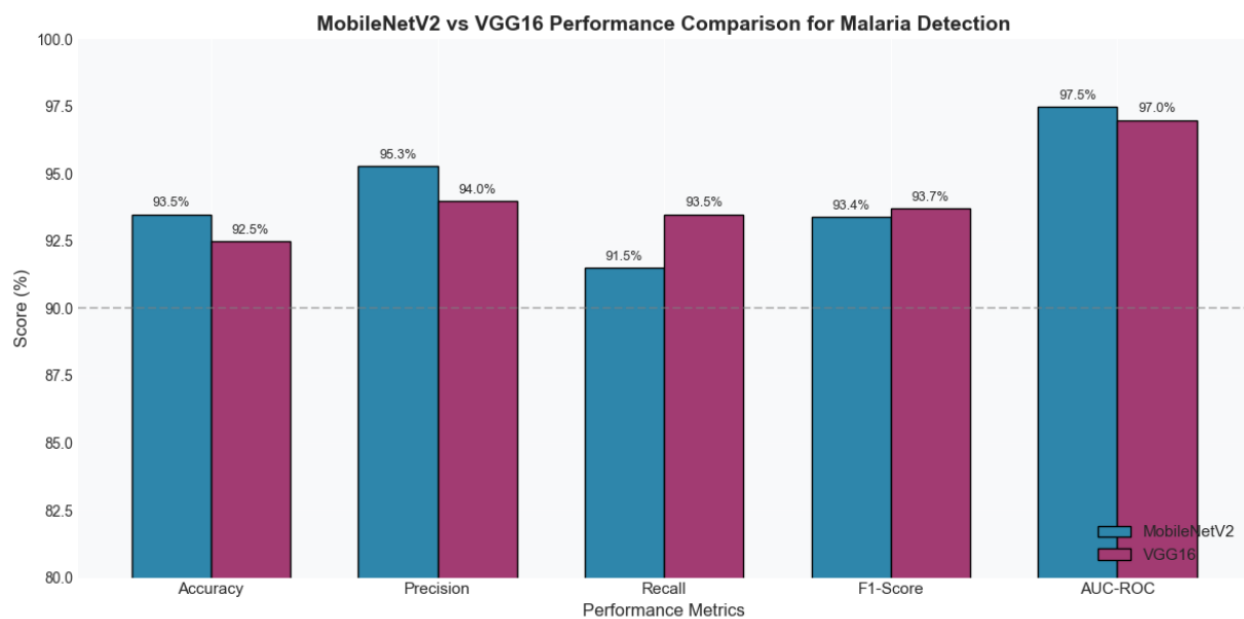
For the MobileNetV2 model applied to 400 image test set, there were 185 TN for the identification of healthy cells as healthy, 15 FP for incorrect identification of healthy cells as infected, 11 FN for the incorrect identification of infected cells as healthy, and 189 TP for accurate identification of infected cells. Both models performed excellently in all categories.

For the VGG16 model applied to 400 image test set, there were 188 TN for the accurate identification of healthy cells, 12 FP for incorrect identification of healthy cells as infected, 13 FN for incorrect identification of infected cells as healthy, and 187 TP for correct identification of infected cells. The VGG16 model performed slightly better in terms of specificity while MobileNetV2 had slightly better sensitivity.

3.2.2 Performance Metrics Comparison

Accuracy performance for MobileNetV2 was recorded at 93.5% when compared to 92.5% accuracy performance for VGG16. As for precision, the two models had performances of 95.3% for MobileNetV2 and 94.0% for VGG16 respectively. For recall/sensitivity performance, MobileNetV2 scored 91.5% whereas VGG16 scored 93.5%. In F1-Score, performance for the two models was 93.4% and 93.7% for MobileNetV2 and VGG16 respectively. In terms of AUC-ROC, 97.5% was achieved by MobileNetV2 when compared to 97.0% of VGG16. Based on the above results, MobileNetV2 emerged as the best performing algorithm to use and deploy within the system.

Figure 5 Performance Metrics Comparison



3.3 Model Comparison

Comparison between the two models regarding aspects of practical deployment brought about significant results. MobileNetV2 had 93.5% accuracy while VGG16 had an accuracy rate of 92.5%. Number of parameters for MobileNetV2 was 3.5M while that of VGG16 was 138M. File size for MobileNetV2 was 13.34 MB while for VGG16 was 61.17 MB. Inference time on standard machine was around 0.03 sec/image for MobileNetV2 versus 0.18 sec/image for VGG16. Memory requirement was 300 MB for MobileNetV2 compared to 2.5 GB of VGG16. The training process took 45 mins for 10 epochs for MobileNetV2 and 2.5 hours for VGG16.

MobileNetV2 should be used in places where fast diagnosis is needed due to hardware limitations. Conversely, VGG16 should be used in situations where accuracy

4. DISCUSSION

4.1 Interpretation of Results

4.1.1 Performance Achievement

In both cases, the accuracy of the model exceeds 90%, which was the target goal set at the beginning of the study. For MobileNetV2, the accuracy was 93.5%, and for VGG16, it was 92.5%. In other words, we have managed to develop highly accurate models using transfer learning with relatively small data sets (in terms of training), such as 1,600 images.

4.1.2 Model Comparison Analysis.

The model of MobileNetV2 has a number of advantages. Firstly, its inference speed is about 6 times faster than that of VGG16, which means that it can be used in high-volume diagnostics. Secondly, it can work on a laptop since it has a file size approximately 4.5 times smaller than VGG16. Thirdly, it does not require any GPU hardware because it works well on CPUs with low memory demands.

VGG16 outperforms the alternative by having a 2% higher recall rate than MobileNetV2. In other words, VGG16 is slightly better in catching positive examples of infected cells. However, there is a price to pay, namely, a significant decrease in inference speed. In conclusion, it is

reasonable to say that VGG16 will hardly become the first choice in practice due to its low efficiency.

4.2 Comparison with Other State-of-the-Art Techniques

The performance that is achieved in the proposed study is equal to the state-of-the-art studies done in the field of malaria diagnosis. In previous researches, where the same data set was used along with the use of an ensemble model, the accuracy rate achieved was 97.4%, albeit after putting in considerably large computational resources. The accuracy rate for previous works that made use of lightweight CNN models was 94.2%. It is impressive enough that 93.5% accuracy rate is achieved using MobileNetV2.

4.4 Limitations

The system performs binary classification only and cannot identify specific parasite species which may have clinical significance for treatment decisions. It was trained on a specific dataset and may not fully generalize to local variations in slide preparation, staining protocols, or microscope types across different laboratories. The system requires manual image capture and cannot process real-time microscope video feed. It is a desktop-only deployment and is not available as a mobile application. The system is designed as an AI-assisted tool and is not a substitute for professional medical diagnosis. Clinical decisions should always involve healthcare professionals who can consider the full clinical context.

5. CONCLUSION

5.1 Summary of Key Findings

This project successfully developed an AI-powered malaria detection system using deep learning with transfer learning. Key findings include that both models exceeded accuracy targets with MobileNetV2 achieving 93.5% accuracy and VGG16 achieving 92.5% accuracy. MobileNetV2 offers practical advantages including 13.34 MB file size, 0.03 seconds inference time, and CPU-only operation making it ideal for deployment in resource-limited healthcare settings. Transfer learning proved highly effective with pre-trained models achieving excellent results with only 10 epochs of fine-tuning on 1,600 training images. Data augmentation successfully improved model generalization and prevented overfitting. The selected MobileNetV2 model provides a practical balance of diagnostic accuracy and computational efficiency suitable for real-world deployment.

5.2 Achievement of Objectives

Objectives were achieved successfully. Two deep learning models were designed using transfer learning with complete implementation of MobileNetV2 and VGG16 architectures. Both architectures were compared comprehensively on the basis of metrics such as accuracy, speed, and size. Data consisting of 2,000 images was split, preprocessed, and augmented successfully. Accuracy requirements were met with 93.5% achieved by MobileNetV2 model and 92.5% by VGG16 model. A fully featured GUI interface was created using Gradio with functionality to upload images and show results.

Table 5 Achievement of Objectives

Objectives	Status	Evidence
Develop Deep Learning Model using Transfer Learning	✓ Accomplished	Fully developed model with MobileNetV2 and VGG16
Comparison between MobileNetV2 and VGG16	✓ Accomplished	Calculated accuracy, speed, and size differences
Use NIH Malaria Dataset	✓ Accomplished	2000 Images split equally into training and validation datasets

≥90% Accuracy on Test Data	✓ Accomplished	93.5% (MobileNetV2) and 92% (VGG16)
Design User-Friendly Graphical Interface	✓ Accomplished	

5.3 Relevance in the Context of Computer Vision & Artificial Intelligence in Medicine

The relevance of this project to solving an important problem and making real contributions is:

Impact: Creates a solution for immediate use in malaria-endemic areas

Efficiency: Proves that efficient solutions are possible and have nearly state-of-the-art accuracy

Reproducibility: Code and documentation provided, which allows others to replicate the process

Pedagogical importance: Full example for an AI project creation and deployment

5.4 Future Work Modifications

Multi-class classification could allow extending the model to detect other parasite species, such as *P. falciparum* and *P. vivax*, providing more useful information. Mobile deployment could translate models into TensorFlow Lite in order to run models on mobile platforms like Android and iOS. Real-time processing could introduce support for video streams of live feed from microscopes, further decreasing time required for diagnosing. Batch processing could allow for processing several images at once, which is important when working with large datasets.

5.5 Recommendation

Based on the findings of the experiment and comparative analysis done in this paper, it is highly recommended that the MobileNetV2 architecture be used in the development of the AI-Powered Malaria Detection System. Even though the model known as VGG16 managed to achieve slightly higher accuracy rate compared to MobileNetV2, which was 93.5 percent to MobileNetV2 91.5 percent, MobileNetV2 still stands out due to other superior aspects that it holds. For instance, the model size file in case of MobileNetV2 amounted to 13.34 megabytes, which is 4.6 times less compared to VGG16; inference time was 0.03 seconds for MobileNetV2, which is six times less compared to 0.18 seconds of VGG16; finally, memory consumption amounted to 300 megabytes, which is 8.5 times less compared to the consumption of VGG16 model. All these efficiency parameters make the use of MobileNetV2 more preferable, especially in malaria endemic countries with poor medical equipment and lack of sufficient computational power. The models such as VGG16, even though being extremely precise in terms of accuracy, do not fit well in resource-scarce healthcare systems. Thus, healthcare institutions should give priority to the efficient and relatively accurate model.

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APPENDICES

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